Research Assistant Test

ERC Really Credible

May 2026

Introduction

This test is designed to reflect the daily tasks of the ERC Really Credible Predoc Team. Its goal is to assess your skills in mathematics, econometrics, and coding, as well as your understanding of the difference-in-differences methods. It also aims to give you a clear sense of what is expected in this predoc position.

Complete **4 out of the 7 exercises**. You must complete **at least one of the two math exercises** (Exercises 1 and 2). You are free to use available resources, provided they comply with academic integrity guidelines.

You should submit this test to `chaisemartin.packages@gmail.com` **by May 19th**. Any format is acceptable, as long as it is readable. Please submit the response as a PDF and include the code file you wrote, if any. Zip these files into a folder named: `LastName_FirstName_RAtest`. We will reach out to you for the outcomes within one week after your submission. Good luck!

- Exercise 1: Super consistent estimator
- Exercise 2: Variance of the estimator of the variance
- Exercise 3: Replicate Burgess et al. (2015)
- Exercise 4: Favorite Econometrics paper
- Exercise 5: Continuous treatment HR estimators
- Exercise 6: User questions
- Exercise 7: Github change

Exercise 1

Do exercise 2 of the PDF `test_RAs_math_1.pdf`.

Exercise 2

Do exercise of the PDF `test_RAs_math_2.pdf`.

Exercise 3: Implement `did_multiplegt_dyn` with Burgess et al. (2015)

In this exercise, you will use the command `did_multiplegt_dyn` to replicate the main two-way fixed effects specification of the study *The Value of Democracy: Evidence from Road Building in Kenya* by Burgess et al. (2015). This paper studies ethnic favoritism in public resource allocation in Kenya.

Burgess et al. (2015) find that in periods of autocracy, coethnic districts (i.e., districts in which the dominant ethnicity matches that of the president in office) receive twice as much expenditure on road building and have five times the length of road built.

Your goal is to estimate dynamic treatment effects of sharing the same ethnicity with the serving president on road building (investment) with `did_multiplegt_dyn`.

Data Description

The dataset is a district-year panel covering all Kenyan districts from 1963 to 2011.

- `distnum`: A unique numeric identifier for each district.
- `year`: Calendar year of observation.
- `exp_dens_share`: Share of road expenditure received by a district (out of the total national road development budget that year), divided by the population share of the district in the national population (in 1962).
- `president`: Indicator variable equal to 1 if the district is coethnic with the serving president.
- `pop1962`: District population in 1962.
- `area`: District size in square km.
- `urbrate1962`: Urbanization rate in 1962.
- `earnings`: District formal total earnings in 1966.

- `wage_employment`: Formal employment in 1963.
- `value_cashcrops`: Value of cash crop exports in 1965.

Benchmark Specification

The benchmark regression considered by Burgess et al. (2015) is the following (Table 1, Panel A):

$$road_{dt} = \gamma_d + \alpha_t + \beta(coethnic\ district_{dt}) + \theta(\mathbf{X}_{d1963} \times [t - 1963]) + u_{dt}$$

Instructions

You may use either the `did_multiplegt_dyn` command in Stata or the equivalent R package. You can consult the respective help files for options of interest and guidance on interpretation (e.g., Stata, R) to answer the following questions:

1. Estimate the dynamic treatment effects of being coethnic with the serving president on road expenditure. Briefly explain your choices for key options such as dynamic effects, placebos, and clustering. Interpret the estimated coefficients and the event-study graph.
2. The authors test the robustness of their main result by sequentially adding a series of controls for demography and economics activity, as presented in Table 1, Column 2-4. Can similar robustness checks be implemented using `did_multiplegt_dyn`? If so, please provide your code and the estimation results. If not, please explain why.
3. Consulting Web Appendix 1.6 of de Chaisemartin and D'Haultfoeuille (2024), is it possible to estimate the effects of switching out of coethnicity with the serving president using `did_multiplegt_dyn` with the method described? If so, please provide your code and the estimation results. If not, please explain why.

Exercise 4

This exercise evaluates how you reason through an open-ended econometrics and software problem, and how you use an AI assistant, such as ChatGPT, Claude, Gemini, or Mistral, to make progress on it. You must submit the full transcript of your conversation with the AI assistant, together with a short write-up summarizing your conclusions.

Problem

`did_multiplegt_dyn` currently computes standard errors through a clustered bootstrap when treatment is continuous. This procedure can be time-consuming because, in each bootstrap iteration, the command must resample the full dataset, run the estimation, and then restore the original data before the next iteration.

One way to improve computational performance is to recognize that the bootstrap can be reformulated as a weighting problem. Each bootstrap iteration can be represented as a random vector of weights, which is then used in the estimation. Under this approach, performing the bootstrap is equivalent to repeatedly running the estimation procedure using different random weighting vectors.

Your task is to code a routine that computes bootstrap standard errors using random weighting vectors. Then, analyze how this weighting procedure affects the main options of the command, including controls, linear trends, parametric trends, and weights, as well as the final calculation of the ATE, placebos, and dynamic effects.

Exercise 5: Continuous treatment HR estimators

You should read the following two working papers:

- Chaisemartin, D'Haultfœuille, Pasquier, Sow, Vazquez-Bare (2025)
- Callaway, Goodman-Bacon, Sant'Anna (2024)

Compare their contributions and discuss the meaningful differences between them.

Exercise 6: User Questions for ChaisemartinPackages

Here are 4 real (anonymized) questions we received from `chaisemartinPackages` users. You should answer each of them by either referring them to proper resources or explaining how to properly use the commands.

1. **Question 1:**

   Hi,

   I hope this email finds you well. My name is xxxxx and I am a final-year Economics undergraduate at the University of xxxxxx. I am writing my dissertation on the UK's Levelling Up policy and I want to use `did_multiplegt_dyn` in Stata. My problem is that I have 4 different treatment variables for the 4 different pots of funding. The first pot of funding was in 2021, the second in 2022, the third in 2023, and the fourth in 2024. I would like to see the effect of each separate funding pot. Is this possible, and if so, what is the correct code? Please let me know if you need any other information.

   Best wishes,
   xxxxxx

2. **Question 2:**

   Hi,

I am using the `did_multiplegt_dyn` command, and I am not sure how to obtain the number of observations used for the graph. For later presentation of the results, I only want to show the graph and the corresponding number of observations. I would be very grateful for your help in this matter.

Best,
xxxxx (PhD Student at xxxx)

3. **Question 3:**

   Hello,

   Thank you for the excellent package in R. I am trying to estimate the effect of a treatment on prices of certain products. The data is a panel consisting of `product_IDs` observed every month for 60 months. The standard model below, including a linear month variable, runs fine.

   ```
   mod_dCDH24 = did_multiplegt_dyn(
     final_sample, 'log_price', 'prod_ID',
     'linear_month', 'new_treatment',
     effects = 12,           % no. of post-treatment periods
     placebo = 12,           % no. of pre-treatment periods
     cluster = 'prod_ID',
   )
   ```

   However, I also wish to account for seasonal effects by month (e.g., because we know there are periods where sales result in heavy discounting). From reading the documentation and issues on GitHub, it seems it is not possible to include dummy variables as controls. Is there a way to control for monthly seasonality at all in this model?

   Thank you in advance for your help.

   Kind regards,
   xxxxxx

4. **Question 4:**

   Hi,

   I have a question about `did_multiplegt_dyn`. Would it be possible to add more fixed effects rather than just "group" and "year"? It would be a variable that only varies for treated units and after the main treatment (a different policy that could contaminate my effects and always happened later, but at different horizons).

   From my reading of the paper and the help manual, controlling for a variable that varies over time only after treatment is different from absorbing another fixed effect. I also know the paper for multiple treatments for the static DiD estimator, but I would like to estimate the dynamic version and simply add the fixed effect of whether and when the groups also got the second policy for a first pass.

   Thank you.
   Best,
   xxxxxx (PhD student in Economics at xxxx)

Exercise 7: Github change

Please review the following GitHub commit made by the team. Summarize the purpose of the change and, if relevant, identify the initial issue it addresses. Your explanation should be concise yet technically detailed. Some modifications aim to enhance user experience, while others introduce new econometric tools. For the latter, clearly explain the implemented method, including formulas where appropriate.

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